My Journey

ESME School of Engineering, Paris 2022 - 2027
M.Sc. in Engineering, Generalist specialized in Bio-Mechatronics

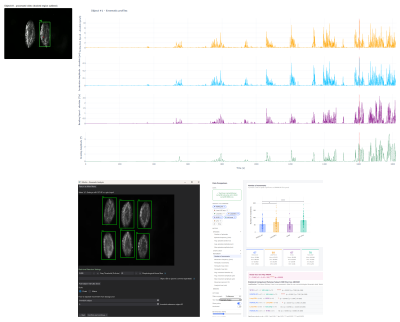
CNRS @ESPCI PSL University (Brain Plasticity Lab) 2026
Computer Vision R&D Intern, Video processing & Kinematic tracking

Universiti Putra Malaysia (UPM) 2025 - 2026
Academic Exchange, Electrical & Electronics Engineering

Danfoss Power Solutions 2025
Manufacturing & Quality Operator, Fluid Power Systems, CNC

ESME Club & FabLab 2022 - 2025
FabLab Administrator & Vice-President, CNC, SLA, FDM & management of hundreds of students

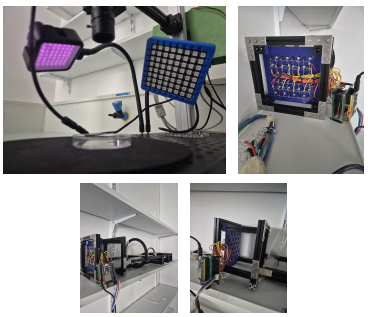
Projects



KiAnSo
September 2026

Kinematic Analysis Software · Python, OpenCV, Computer Vision, Kinematics

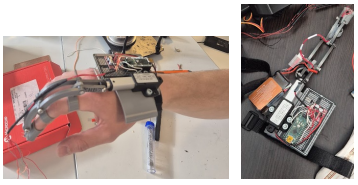
Kinematic analysis software developed at the Brain Plasticity Lab (CNRS / ESPCI-PSL) for behavioral neuroscience on Drosophila. Built from scratch in Python to replace a legacy tool, it models embryonic spasms and adult locomotion using Farnebäck dense optical flow, robust signal filtering (Hampel and Savitzky-Golay), and batch processing, delivering a 500% throughput increase and saving over 45 hours per experimental run.



OptoStim
September 2026

Optogenetics Bench & Synchronized Video Acquisition · Python, Flask, Hardware / IoT

Automated optogenetic photostimulation and high-speed video acquisition platform designed for Drosophila behavioral neuroscience at the Brain Plasticity Lab (CNRS / ESPCI-PSL). Engineered the full software architecture in Python and Flask, integrating programmable pulse trains and automated pre/post-stimulus video capture synchronization. Directly interfaced with embedded microcontrollers and coupled to the KiAnSo analysis suite for millisecond-accurate behavioral response tracking.



AILAPS
May 2026

Active Hand Exoskeleton · Bio-Mechanics, SolidWorks, Embedded C, Actuation

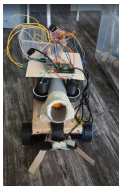
Motorized active hand exoskeleton prototype developed for rehabilitation and assistance with everyday muscular weakness. Integrates custom compliant linkages, embedded driver electronics, and deterministic actuation.



Cuicui
June 2023

Upper-Limb Pediatric Prosthetic Device · Biomechanics, Prototyping, Real Case, 3D Printing

Custom mechanical prosthetic device developed for a young child with congenital limb agenesis who wanted to ride a bike. Designed from direct alginate and plaster limb casting, prioritizing practical usability for the girl, lightweight comfort, and reliable mechanical handlebar control without external power.




Scotchouille

June 2025

Autonomous Mobile Rover · Python, Computer Vision

Tournament-winning autonomous competition robot. Equipped with differential drive kinematics, live computer vision ball detection, dynamic trajectory planning, and a high-velocity projectile launching mechanism.



Louya

December 2016

Custom Creality CR-10 · Prototyping, Materials, Firmware, DFM/DFA

Heavily modified Creality CR-10 upgraded over the years with a custom direct-drive setup, all-metal hotend, and tuned firmware. To be honest, I end up breaking it almost every print trying to over-optimize things that were working just fine, which is never a good idea, but definitely where the real hands-on learning happens.

Experiments

Can 8 equations replace 80 years of PID control? Not quite.

Control Theory & Robotics · PID Controller, Neural Network (MLP), Hybrid Control

For the past two years in engineering school, I've had PID controllers drilled into my head in virtually every control systems class. I even traveled halfway around the world to Malaysia, studying alternative methods for model identification and parameter tuning. PID is everywhere and over 95% of industrial automation relies on it, but it made me wonder: why don't engineering curriculums show emerging machine learning techniques, like compact Multilayer Perceptrons, for state correction and complex nonlinear systems?

To investigate, I simulated a quadrotor carrying a suspended payload, a notoriously underactuated nonlinear system where moving the drone excites violent pendular swings. Around small angles (under 5 degrees), linear approximations hold, and a tuned PID provides near-optimal damping without any machine learning. Above 20 degrees of tilt, nonlinear centrifugal and Coriolis couplings dominate. Fixed linear gains face an impossible tradeoff: tune aggressively to suppress payload oscillations, and the motors chatter and saturate; tune softly for stability, and the payload swings wildly past the target (27.3 cm average miss).

Instead of running a heavy Nonlinear Model Predictive Controller (NMPC) at 200 Hz, I trained a compact 2-layer MLP offline via behavioral cloning from an optimal trajectory solver, compiling it into just 106 parameters (8 linear algebra equations in C). Evaluating in under 2.5 microseconds on an ARM Cortex-M4, it automatically counter-pitches half a period before arrival to cancel angular momentum, cutting tracking error by 43% (15.5 cm average miss) with zero oscillatory ringing.

However, pure neural control has a blind spot: sustained disturbances. By the Francis-Wonham Internal Model Principle, rejecting a constant external force (like a 1.4 N crosswind) mathematically requires an integrator. Because a feedforward MLP has no internal state, it suffers a permanent steady-state drift under wind. The winning architecture is hybrid: a classical PID integral baseline guaranteeing zero steady-state error, augmented by a lightweight neural policy to eliminate nonlinear swing.

Can we predict the outcome of a coin toss in mid-air?

State Estimation & Bayesian Filtering · C++, Orocos BFL, Extended Kalman Filter, Particle Filter, OpenCV

Lately, I've been diving into state estimation and the Orocos ecosystem in ROS. To challenge myself beyond standard 2D robot localization, I built a 3D coin-toss tracking simulator in C++ using OpenCV and Orocos BFL (Bayesian Filtering Library). A coin toss isn't random magic, but pure deterministic physics: if you can estimate the state vector in the first milliseconds of flight, you can calculate the landing face before impact.

I set up a 4D state vector $X = [z, \dot{z}, \theta, \omega]$. Under a simulated 30 Hz camera with Gaussian measurement noise ($R = 0.25\,\mathrm{px}^2$), I benchmarked two BFL estimators: an EKF (with hand-calculated Jacobians) and an SMC Particle Filter (about 600 particles). The EKF executed in just 0.04 ms per step, locking onto the 37.7 rad/s spin in 5 frames ($\sim$166 ms of flight) and dropping estimation error to $< 1.2\%$ while tracking covariance fell from 10.0 to 0.04. The Particle Filter took 4.2 ms per step; though slower, it handled temporary face occlusions by tracking multi-modal probability distributions before collapsing them.

Testing the camera's observability limits revealed a fascinating control theory trap. At a 45° viewing angle, EKF prediction accuracy reached 96.4% after 200 ms. Whereas at a 90° angle (looking directly at the coin's thin edge), the Jacobian became ill-conditioned, covariance exploded from 0.01 to 4.2, and prediction accuracy collapsed to 51.5%. Flat landings yielded 98% confidence, while edge landings dropped EKF confidence to 54%. The filter literally knows when it doesn't know.